EECS 498/CSE 598: Zero-Knowledge Proofs
Winter 2026
Lecture 9

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Agenda for this lecture

- Announcements
- The Delphian PIOP for R1CS-Sat
- Soundness of Delphian as a PIOP
- Compiling (P)IOPs to arguments (of knowledge)
- Sigma protocols

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Announcements

- Project 2 is online
- Autograder is working
- Due date: 2/20
- How about that Super Bowl? I can’t believe $TEAM_NAME won!

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Delphian PIOP

(first phase)

$\widetilde{eq}: \{0,1\}^{\log n} \times \{0,1\}^{\log n} \to \{0,1\}$

$\mathcal{P}(A, B, C, x, w)$		$\mathcal{V}(A, B, C, x)$
compute multilinear extension $\tilde{w}$ of w		
	$\xrightarrow{\mathcal{O}_{\tilde{w}}}$	
		$r \leftarrow\$ \mathbb{F}^{\log n}$
	$\xleftarrow{r}$	
compute $h(x) = \widetilde{\mathsf{eq}}(r, x) g_{\text{res}}(x)$		
	$\xrightarrow{\mathcal{O}_h}$	
	$\xleftrightarrow{\mathsf{SCPIOP}[\mathcal{P}_{sc}(h, 0) \leftrightarrow \mathcal{V}_{sc}^{\mathcal{O}_h}(0)]}$ $\mathcal{V}_{sc} \to r' = (r'_1, \ldots, r'_{\log n}), H'$	
for $M \in \{A, B, C\}$,		$e \leftarrow \widetilde{\mathsf{eq}}(r, r')$; $G' \leftarrow H'/e$
$p_M(x) = \tilde{f}_M(r', x)\tilde{z}(x)$		[Note G' is claimed to be $g_{\text{res}}(r')$]
$v_M = \sum_{x \in \{0,1\}^{\log n}} p_M(x)$		
	$\xrightarrow{v_A, v_B, v_C, \mathcal{O}_{p_A}, \mathcal{O}_{p_B}, \mathcal{O}_{p_C}}$	
		Check $v_A * v_B - v_C = G'$

$h(r'_1, \ldots, r'_{\log n}) = H'$

$g_{res}(X) = \left(\sum \tilde{A}(X, j)\tilde{z}(j)\right)\left(\sum \tilde{B}(X, j)\tilde{z}(j)\right) - \left(\sum \tilde{C}(X, j)\tilde{z}(j)\right)$

$\forall X,\ g_{res}(X) = 0$

$h(X) = \widetilde{eq}(\tau, X) g_{res}(X)$

$\sum h(X) = 0$

Delphian PIOP

(second phase)

$$p_M(r', y) := \tilde{M}(r', y)\,\tilde{Z}(y) \quad \text{for } M = A, B, C$$

$$g_{res}(r') = \left(\sum p_A(r', y)\right)\left(\sum p_B(r', y)\right) - \left(\sum p_C(r', y)\right)$$

$$G' = v_A v_B - v_C$$

Delphian PIOP

(second phase)

$p_M(r''_M) = \tilde{M}(r', r''_M)\tilde{Z}(r''_M)$

Complete for $M \in \{A, B, C\}$:

$$\mathsf{SCPIOP}[\mathcal{P}_{sc}(p_M, v_M) \leftrightarrow \mathcal{V}_{sc}^{\mathcal{O}_{p_M}}(v_M)]$$

$$\mathcal{V}_{sc} \to r''_M = (r''_{M,1}, \ldots, r''_{M,\log n}), H''$$

$$\text{Query } \mathcal{O}_{\tilde{w}}(r''_M[:-1])$$

$$\mathcal{V} \text{ gets } w_M \leftarrow \tilde{w}(r''_M[:-1])$$

$$z_M = (1 - r''_M[-1])\tilde{x}(r''_M[:-1]) + r''_M[-1]w_M$$

$$e_M = \tilde{f}_M(r', r''_M)z_M$$

$$\text{Check } H'' = e_M$$

Final checks

If any subprotocols output $\perp$ or checks fail, output $\perp$, else True

$n = 2^\ell$

$$\tilde{Z}(Y_1, \ldots, Y_\ell) = (1 - Y_\ell)\tilde{x}(Y_1, \ldots, Y_{\ell-1}) + Y_\ell \tilde{w}(Y_1, \ldots, Y_{\ell-1})$$

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Soundness of the Delphian PIOP

Theorem 2. *Let $n = 2^\ell$ and $A, B, C \in \mathbb{F}^{n\times n}$ be an R1CS over $\mathbb{F}$ and take $x \in \mathbb{F}^{n/2} \notin \mathcal{L}_{A,B,C,n,\mathbb{F}}$. Let $\mathbb{V}_D$ be the Delphian PIOP verifier. For any malicious PIOP prover $\mathbb{P}^*$,*

- check $\forall g_{res}$ using $\sum h$
- Sumcheck security as PIOP
- PIOP model

$$\Pr[\mathrm{out}_{\mathbb{V}_D}[\mathbb{P}^*(A,B,C,n,x) \leftrightarrow \mathbb{V}_D(A,B,C,x)] = 1] \leq \frac{10\ell}{|\mathbb{F}|}$$

Proof: $x \notin \mathcal{L}$, no matter what $\tilde{w}$ is in oracle

$g_{res}(x) \neq 0$ so $\Pr[\sum h = 0] \leq \frac{\ell}{|\mathbb{F}|}$

To pass first phase, $\mathbb{P}^*$ breaks SC-PIOP

$\Pr[\mathrm{out}_{\mathbb{V}_{SC_1}}[\mathbb{P}^* \leftrightarrow \mathbb{V}_{SC_1}]$ is true claim about $h] \leq \frac{3\ell}{|\mathbb{F}|}$

If $h(r') \neq H'$, at least one of $U_\alpha, V_\beta, V_\gamma$ wrong

Soundness of the Delphian PIOP

Theorem 2. *Let $n = 2^\ell$ and $A, B, C \in \mathbb{F}^{n\times n}$ be an R1CS over $\mathbb{F}$ and take $x \in \mathbb{F}^{n/2} \notin \mathcal{L}_{A,B,C,n,\mathbb{F}}$. Let $\mathbb{V}_D$ be the Delphian PIOP verifier. For any malicious PIOP prover $\mathbb{P}^*$,*

$$\Pr[\mathsf{out}_{\mathbb{V}_D}[\mathbb{P}^*(A,B,C,n,x) \leftrightarrow \mathbb{V}_D(A,B,C,x)] = 1] \leq \frac{10\ell}{|\mathbb{F}|}$$

$\Pr[\mathsf{out}_{\mathbb{V}_{SC_\mu}}[\mathbb{P}^*_{SC_\mu}(\mu, v_\mu) \leftrightarrow \mathbb{V}_{SC_\mu}(v_\mu)]$ is true claim$] \leq \frac{2\ell}{|\mathbb{F}|}$

<u>Claim:</u> If $P_\mu(r''_\mu) \neq H''_\mu$ for some μ, $\mathbb{P}^*$ (by construction + PIOP restrictions)

Union bound over phases 1+2

$\boxtimes$

Pop quiz

At the end of the first phase, the verifier checks that $v_A v_B - v_C = G'$.
Why is this check important? Explain in your own words.

Pop quiz

At the end of the first phase, the verifier checks that $v_A v_B - v_C = G'$.
Why is this check important? Explain in your own words.

<u>Exercise</u>: Construct a cheating prover that breaks soundness w.h.p. if the verifier does not do this check.

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Knowledge soundness

Soundness $\not\Rightarrow$ Knowledge Soundness

If $\mathbb{P}$ convinces $\mathbb{V}$ w.p. ε,
then $\exists$ extractor E that can
interact with $\mathbb{P}$ to get w
w.p. "almost" ε

Formalizing knowledge soundness is pretty difficult… we could probably spend the rest of the semester studying just this question…

Knowledge soundness

Definition. *An interactive proof* $\mathsf{IP} = \langle \mathcal{P}, \mathcal{V} \rangle$ *for the language* $\mathcal{L}$ *is a proof of knowledge for* $\mathcal{L}$ *with* knowledge error κ *if there exists an expected PPT* extractor E *such that for any prover* $\mathcal{P}^*$ *and* $x \in \mathcal{L}$,

$$\Pr[E^{\mathcal{P}^*}(x) \in \mathcal{R}_{\mathcal{L}}(x)] \geq \Pr[\mathsf{out}_{\mathcal{V}}[\mathcal{P}^*(x) \leftrightarrow \mathcal{V}(x)] = 1] - \kappa \ .$$

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Compiling PIOPs to arguments

Compiling PIOPs to arguments

Claim. *Let* $\mathsf{PIOP} = \langle \mathcal{P}_{\mathsf{PIOP}}, \mathcal{V}_{\mathsf{PIOP}} \rangle$ *be a polynomial interactive oracle proof for* $\mathcal{L}$ *with degree and variable bounds* $(d_1, v_1), \ldots, (d_k, v_k)$. *Let* $\mathsf{PC} = (\mathsf{Setup}, \mathsf{Com}, \mathsf{Open})$ *be a polynomial commitment scheme. Let* $\mathsf{Arg}[\mathsf{PIOP}, \mathsf{PC}]$ *be the argument obtained from the "oracle-to-PC" compiler. If* $\mathsf{PC.Com}$ *is binding,* $\mathsf{PC.Open}$ *is a proof of knowledge, and* PIOP *is sound, then* $\mathsf{Arg}[\mathsf{PIOP}, \mathsf{PC}]$ *is a sound argument for* $\mathcal{L}$.

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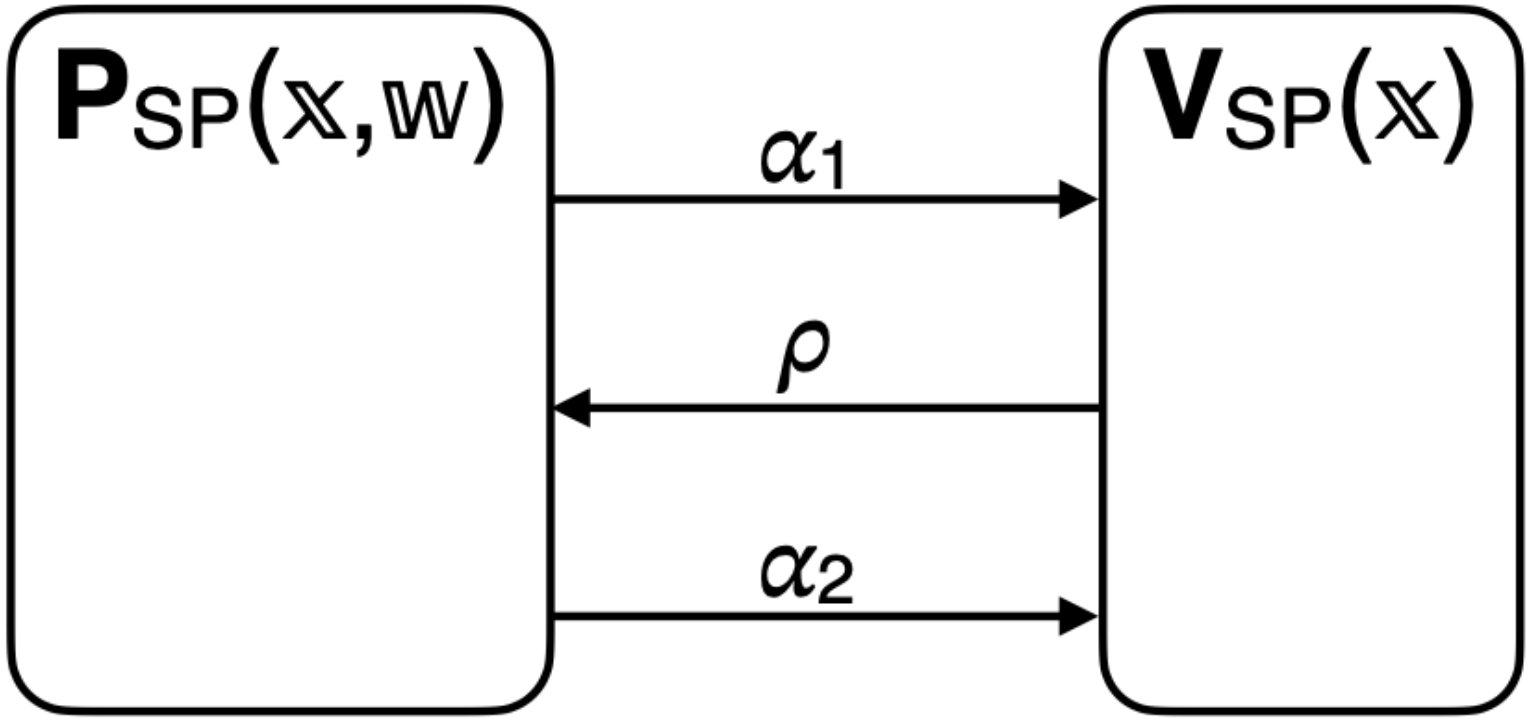
Sigma protocols